

GAMMA REMOTE SENSING

NextGenCarbon
Monitoring Earth's Carbon Balance

An early assessment of BIOMASS data in support of global mapping of AGB

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19th EEBIOMASS workshop on "BIOMASS mission Cal/Val activities"

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Role of the BIOMASS mission

- The start of operations of the BIOMASS satellite provides a novel data stream that in principle shall improve the estimation of AGB from space compared to current activities at GAMMA, e.g., CCI Biomass.
- Complementing current mapping activities foresees two options
    1. Ingest BIOMASS Level 1 data or derived products (backscatter, InSAR, PolInSAR, TomoSAR) in a biomass estimation framework, complementing retrievals at C- and L-band
    2. Ingest BIOMASS Level 2 data in an AGB post-processing framework (e.g., bias-correct existing maps)

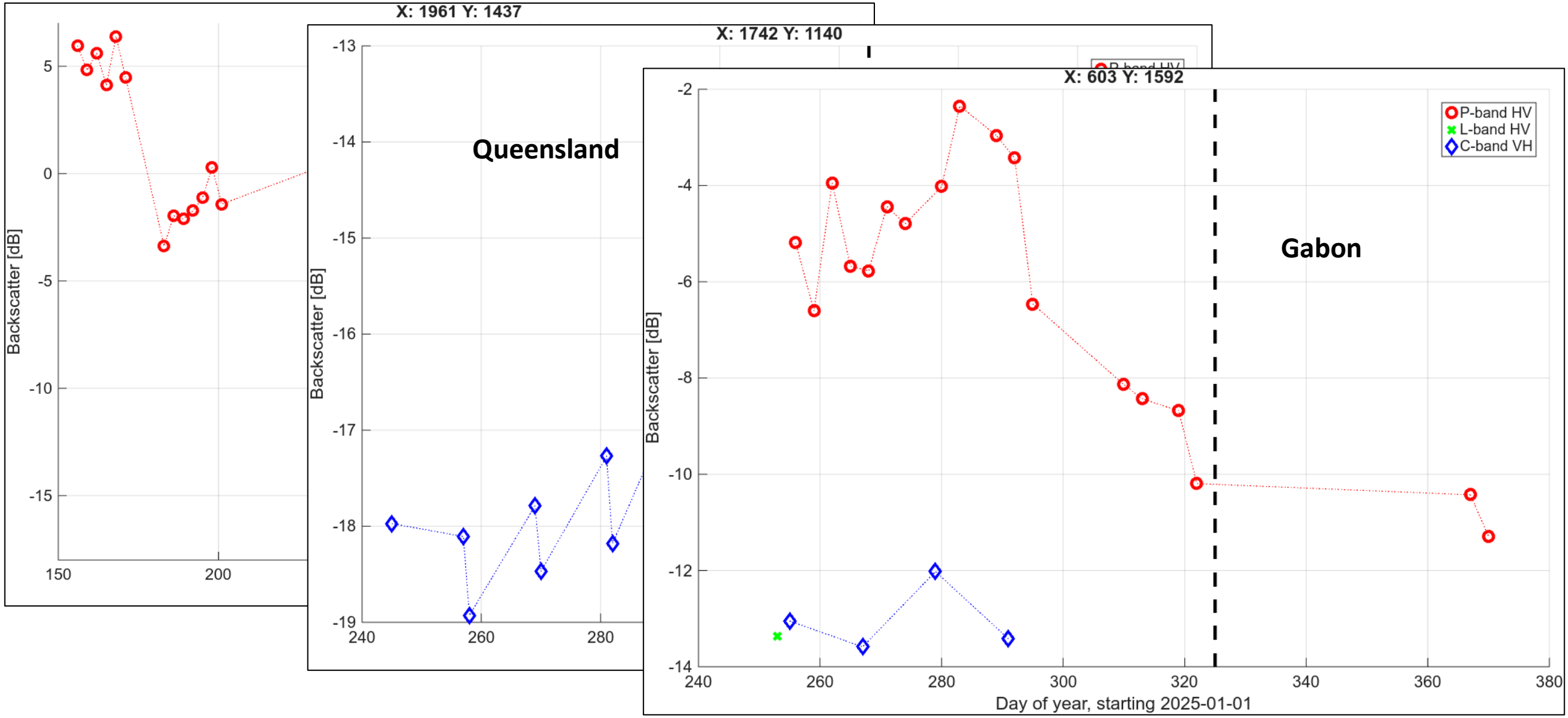
- BIOMASS provides data with temporal and spatial features that are mostly unknown to the data user communities → both pathways need to be explored to have a correct understanding of how P-band observations contribute to wall-to-wall quantification of carbon stored in vegetation.

Scope of this work

- Data from the Cal/Val phase and first data from the Tomo Phase are relevant to understand the contribution of BIOMASS to the quantification of AGB from space.
- At the early stage of the mission, we pursue signature analyses of BIOMASS standard products

1. Investigate temporal stability and quality of the BIOMASS backscatter values using IOC data
2. Level 2 products investigation:
    1. Forest Height stability and correlation with ALS derived AGB
    2. Ground Notched backscatter comparison to ALS derived AGB and L1a products

Temporal profiles



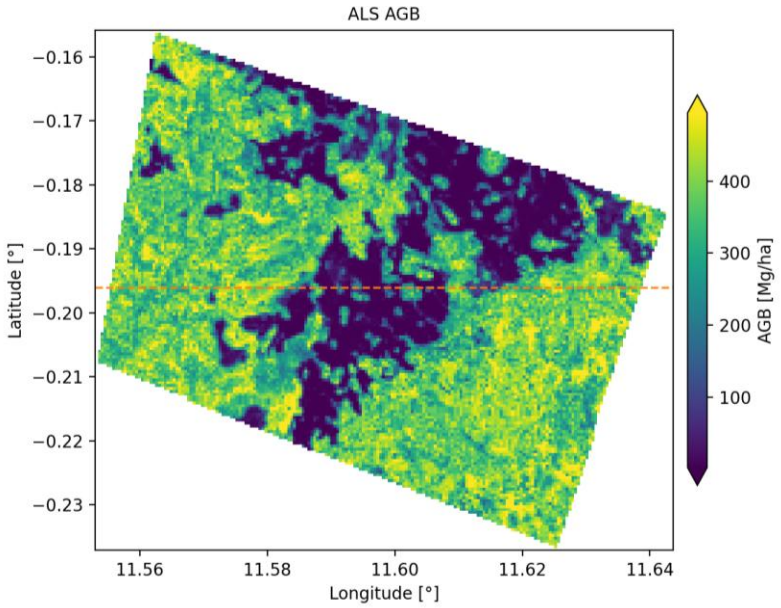
Recalibration of IOC data necessary to understand the signals in time

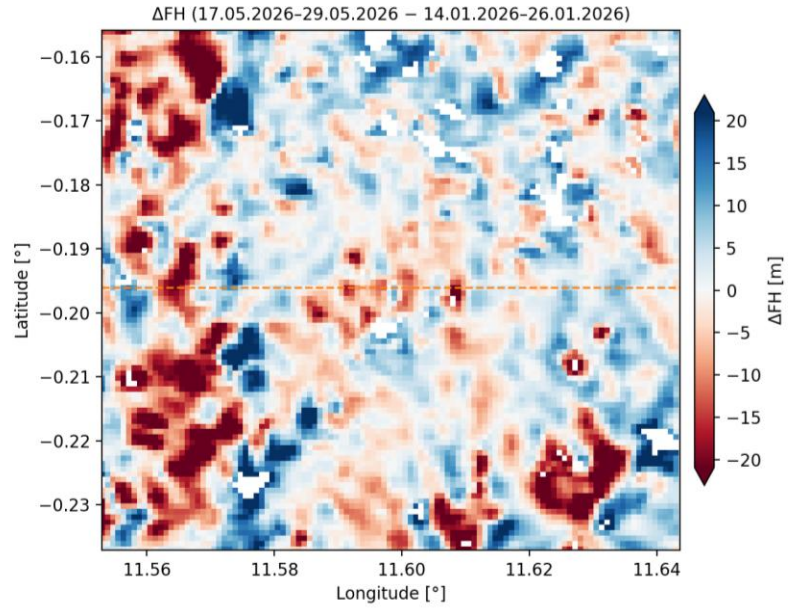
Level 2 Forest Height products

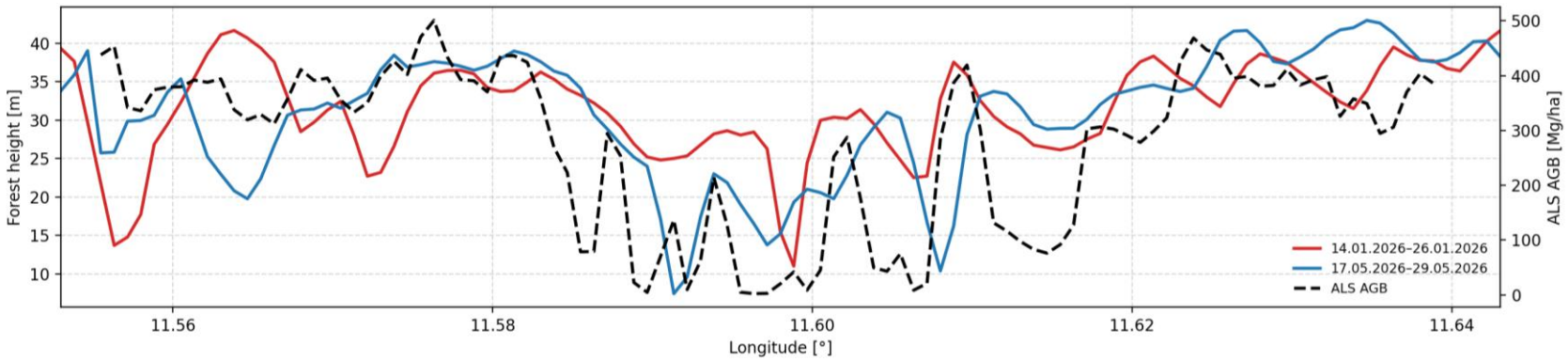
ALS derived AGB and Forest Height difference in Lope, Gabon (Evergreen Tropical Forest, 2016)

Understand the effect of algorithm parameters (data masks, topographic correction, etc.) by comparing successive estimates of FH (January, May).

Currently low correlation (r=0.5) future versions should improve L2 FH product stability and accuracy.





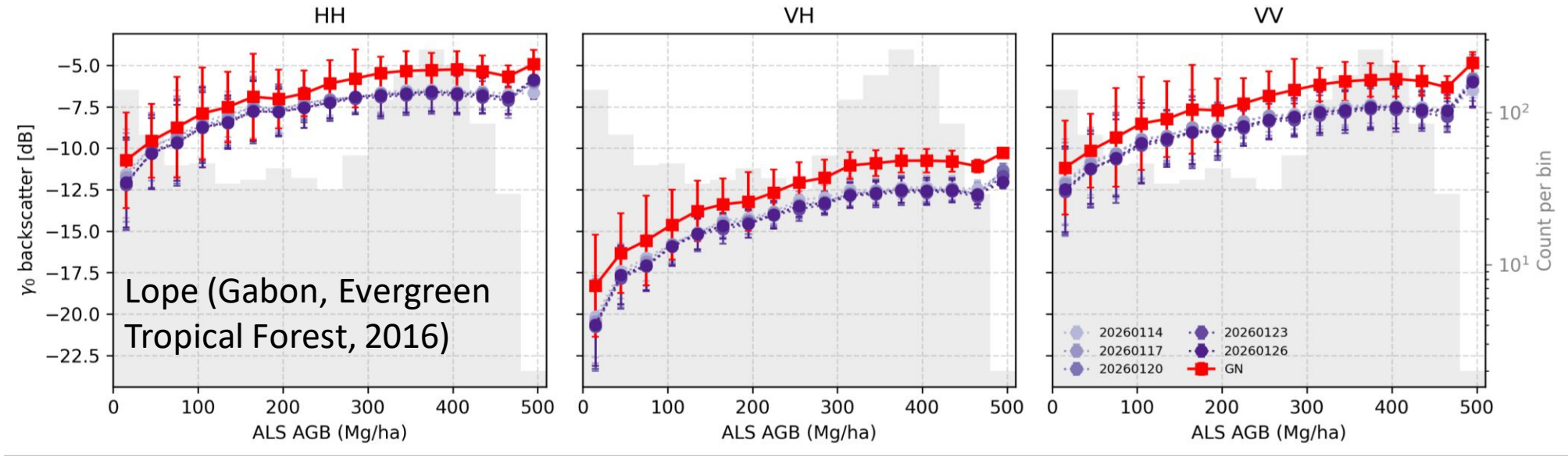


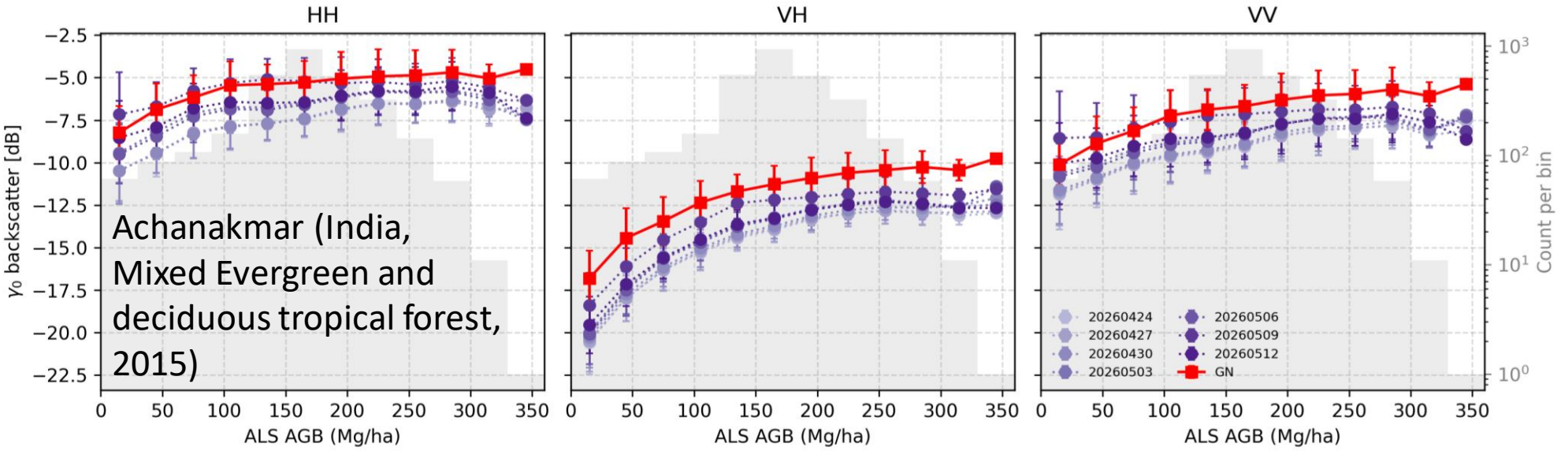
Level 2 Ground Notch product

GN (simplified tomography removing ground contribution in the backscatter), compared to ALS derived AGB

One GN image created per stack (red), L1A backscatter from same stack (blue)

Most corrections appear close to topographic feature or forest edges, limited impact on the sensitivity to AGB variations.





Conclusions and suggestions

- IOC represent a unique chance to investigate P-band signals in time. Fully calibrated dataset is recommended to understand the signals in time
- Now in the TOM phase of acquisitions, available data amount increases every week:
  - Quality of the Level 1 products in line with expectation
  - RFI and ionospheric contaminations still occur
- Level 2 FH does not appear to be reliable now, updated algorithms should improve this.
- Level 2 GN does not provide clear sensitivity improvements over Level 1 backscatter
- Further investigation will look at more recent ALS derived data (GEO-TREES) and differences with the L and C-band backscatter